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Liquid chemical supply module and substrate processing apparatus including the same

Abstract

Provided is a liquid chemical supply module and a substrate processing apparatus including the same, the liquid chemical supply module including a main supplier for supplying a liquid chemical from a liquid chemical tank to at least one substrate processing apparatus, and at least one distribution supplier for connecting between the main supplier and any one substrate processing apparatus to distribute the liquid chemical supplied through the main supplier to the any one substrate processing apparatus, wherein the distribution supplier includes a distribution line branched from a main line of the main supplier, a first circulation line branched from the distribution line and connecting between the distribution line and the main line, and a second circulation line branched from the distribution line connecting between the distribution line and the main line.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims the benefit of Korean Patent Application No. 10-2022-0147337, filed on Nov. 7, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a liquid chemical supply module and a substrate processing apparatus including the same and, more particularly, to a liquid chemical supply module capable of supplying a liquid chemical used to process a substrate to a substrate processing apparatus, and a substrate processing apparatus including the same.

2. Description of the Related Art

(3) Various processes such as photolithography, etching, ashing, and deposition are performed to manufacture semiconductor devices or liquid crystal displays. In these processes, due to the miniaturization of circuit patterns on a substrate, device characteristics and production yield are more sensitively affected by particles and contaminants. As such, a cleaning process for removing the particles and contaminants is performed before or after each process. A plurality of chemical supplying, rinsing, and drying processes may be sequentially performed as the cleaning process in a chamber, and contaminants or particles and fumes which are produced in the cleaning process may be exhausted to the outside by inducing the downflow of clean air in the chamber during the cleaning process.

(4) When the substrate is cleaned, various types of liquid chemicals may be used to remove the contaminants on the substrate, and each liquid chemical may be ejected onto the substrate through an ejection nozzle positioned above the substrate. The liquid chemical to be ejected through the ejection nozzle to clean the substrate may be controlled to a flow rate suitable for process conditions by a liquid chemical supply module connected to a liquid chemical tank, and then supplied to the ejection nozzle of a substrate processing apparatus.

(5) The liquid chemical supply module for supplying the liquid chemical from the liquid chemical tank to the substrate processing apparatus may be connected to a plurality of substrate processing apparatuses to supply the liquid chemical thereto and, in this case, a back pressure valve (BPV) may be mounted on a main line connected to the liquid chemical tank so as to maintain a pressure of the liquid chemical in the main line at a constant pressure.

(6) However, according to the existing liquid chemical supply module, although the liquid chemical is ejected at the same flow rate through the main line, an opening ratio of a flow control valve (FCV) needs to be controlled differently for a chamber of each substrate processing apparatus. For example, when a plurality of chambers are used, a front pressure of the liquid chemical supplied through the main line may be reduced and thus the opening ratio of the FCV may be changed to maintain the ejection rate of the liquid chemical supplied to each chamber. At this time, a dead zone may be reduced due to the change in the opening ratio of the FCV and particles stagnant in the FCV may also be ejected onto the substrate of the substrate processing apparatus. In addition, by setting the opening ratio of the FCV differently for each substrate processing apparatus, equipment and chamber performance deviations may be caused.

SUMMARY OF THE INVENTION

(7) The present invention provides a liquid chemical supply module capable of always maintaining a constant pressure of a liquid chemical, minimizing operation of a flow control value (FCV) during ejection of the liquid chemical, and thus stably maintaining particle performance by mounting a relief valve at a front end of the FCV of every substrate processing apparatus connected to the liquid chemical supply module, and a substrate processing apparatus including the liquid chemical supply module. However, the above description is an example, and the scope of the present invention is not limited thereto.

(8) According to an aspect of the present invention, there is provided a liquid chemical supply module including a main supplier for supplying a liquid chemical from a liquid chemical tank to at least one substrate processing apparatus, and at least one distribution supplier for connecting between the main supplier and any one substrate processing apparatus to distribute the liquid chemical supplied through the main supplier to the any one substrate processing apparatus, wherein

the distribution supplier includes a distribution line branched from a main line of the main supplier and connecting between the main line and the any one substrate processing apparatus, a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line, and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line.
(9) The main supplier may include the main line for supplying the liquid chemical from the liquid chemical tank, and a back pressure valve (BPV) mounted on the main line in front of the distribution line with respect to a direction of the liquid chemical flowing through the main line, to maintain a pressure of the liquid chemical flowing in the main line at a preset certain main pressure.
(10) The main supplier may further include a pressure sensor mounted on the main line between the BPV and the distribution line to sense the pressure of the liquid chemical flowing in the main line behind the BPV.
(11) The distribution line may include a flow control valve (FCV) mounted on the distribution line to control a flow rate of the liquid chemical supplied to the any one substrate processing apparatus.
(12) The distribution line may further include a first on/off valve mounted on the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to open or close the distribution line.
(13) The first circulation line may be branched from the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line.
(14) The first circulation line may include a second on/off valve mounted on the first circulation line to open or close the first circulation line.
(15) The second circulation line may be branched from the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line.
(16) The second circulation line may include a relief valve mounted at a position where the second circulation line is branched from the distribution line, to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line when the pressure of the liquid chemical flowing in the distribution line exceeds a preset certain distribution pressure.
(17) The second circulation line may further include a third on/off valve mounted on the second circulation line to open or close the second circulation line. The distribution line may further include a flowmeter mounted on the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to measure a flow rate of the liquid chemical flowing in the distribution line.
(18) The distribution line may further include a flow controller electrically connected to the FCV and the flowmeter to receive a value of sensing the flow rate of the liquid chemical from the flowmeter and apply a control signal to the FCV.
(19) The flow controller may apply the control signal to the FCV based on a difference between a preset reference flow rate and an actual flow rate of the liquid chemical calculated using the sensed value received from the flowmeter, to control the difference between the reference flow rate and the actual flow rate of the liquid chemical within a preset certain error range.
(20) The main supplier may set the main pressure of the liquid chemical having passed through the BPV and flowing in the main line, to be higher than a distribution pressure of the liquid chemical

distributed through the distribution supplier and supplied to the any one substrate processing apparatus.
(21) The distribution supplier may include a plurality of distribution suppliers mounted at certain intervals along the main line of the main supplier to distribute the liquid chemical supplied through the main supplier, to a plurality of substrate processing apparatuses.
(22) The distribution suppliers may include a first distribution supplier for connecting between the main line and a first substrate processing apparatus, and a second distribution supplier for connecting between the main line and a second substrate processing apparatus behind the first distribution supplier with respect to a direction of the liquid chemical flowing through the main line.
(23) Each of the first and second distribution suppliers may include the distribution line, the first circulation line, and the second circulation line.
(24) The distribution suppliers may further include an nth distribution supplier for connecting between the main line and an nth substrate processing apparatus behind an (n−1)th distribution supplier with respect to the direction of the liquid chemical flowing through the main line.
(25) According to another aspect of the present invention, there is provided a substrate processing apparatus including a chamber including a processing space where a substrate is processed, a substrate support unit mounted in the processing space to support and rotate the substrate, an ejection unit mounted above the substrate support unit in the processing space to eject at least one type of liquid chemical onto the substrate through an ejection nozzle, a liquid chemical supply module for supplying the liquid chemical to the ejection unit, and a collection unit provided in a vessel shape with an open top to surround the substrate support unit and mounted in the processing space to collect the liquid chemical scattered from the substrate when the substrate rotates, wherein the liquid chemical supply module includes a main supplier for supplying the liquid chemical from a liquid chemical tank to the ejection nozzle of the ejection unit, and a distribution supplier for connecting between the main supplier and the ejection nozzle to distribute the liquid chemical supplied through the main supplier to the ejection nozzle of the ejection unit, and wherein the distribution supplier includes a distribution line branched from a main line of the main supplier and connecting between the main line and the ejection nozzle, a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line, and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line.
(26) According to another aspect of the present invention, there is provided a liquid chemical supply module including a main supplier for supplying a liquid chemical from a liquid chemical tank to at least one substrate processing apparatus, and at least one distribution supplier for connecting between the main supplier and any one substrate processing apparatus to distribute the liquid chemical supplied through the main supplier to the any one substrate processing apparatus, wherein the distribution supplier includes a distribution line branched from a main line of the main supplier and connecting between the main line and the any one substrate processing apparatus, a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line, and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line, wherein the main supplier includes the main line for supplying the liquid chemical from the liquid chemical tank, a back pressure valve (BPV) mounted

on the main line in front of the distribution line with respect to a direction of the liquid chemical flowing through the main line, to maintain a pressure of the liquid chemical flowing in the main line at a preset certain main pressure, and a pressure sensor mounted on the main line between the BPV and the distribution line to sense the pressure of the liquid chemical flowing in the main line behind the BPV, wherein the distribution line includes a flow control valve (FCV) mounted on the distribution line to control a flow rate of the liquid chemical supplied to the any one substrate processing apparatus, a flowmeter mounted on the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to measure a flow rate of the liquid chemical flowing in the distribution line, and a first on/off valve mounted on the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to open or close the distribution line, wherein the first circulation line is branched from the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line, and wherein the second circulation line is branched from the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line, and includes a relief valve mounted at a position where the second circulation line is branched from the distribution line, to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line when the pressure of the liquid chemical flowing in the distribution line exceeds a preset certain distribution pressure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other features and advantages of the present invention will become more apparent by describing in detail embodiments thereof with reference to the attached drawings in which:
(2) FIG. **1** is a conceptual view of a liquid chemical supply module according to an embodiment of the present invention;
(3) FIG. **2** is a conceptual view of a portion of a distribution supplier of the liquid chemical supply module of FIG. **1**; and
(4) FIG. **3** is a conceptual view of a substrate processing apparatus including the liquid chemical supply module of FIG. **1**.
DETAILED DESCRIPTION OF THE INVENTION
(5) Hereinafter, the present invention will be described in detail by explaining embodiments of the invention with reference to the attached drawings.
(6) The invention may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the concept of the invention to one of ordinary skill in the art. In the drawings, the thicknesses or sizes of layers are exaggerated for clarity and convenience of explanation.
(7) Embodiments of the invention are described herein with reference to schematic illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, the embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein, but are to include deviations in shapes that result, for example, from manufacturing.
(8) FIG. **1** is a conceptual view of a liquid chemical supply module **1000** according to an

embodiment of the present invention, FIG. **2** is a conceptual view of a portion of a distribution supplier **1200** of the liquid chemical supply module **1000** of FIG. **1**, and FIG. **3** is a conceptual view of a substrate processing apparatus **10** including the liquid chemical supply module **1000** of FIG. **1**.

(9) Initially, as shown in FIG. **1**, the liquid chemical supply module **1000** according to an embodiment of the present invention may mainly include a main supplier **1100** and a plurality of distribution suppliers **1200**.

(10) As shown in FIG. **1**, the main supplier **1100** may supply a liquid chemical from a liquid chemical tank (not shown) to a plurality of substrate processing apparatuses **10**, and the plurality of distribution suppliers **1200** may connect between the main supplier **1100** and first, second, third, fourth, and fifth substrate processing apparatuses **10-1**, **10-2**, **10-3**, **10-4**, and **10-5** to distribute the liquid chemical supplied through the main supplier **1100** to the plurality of substrate processing apparatuses **10**.

(11) Specifically, the main supplier **1100** may include a main line **1110** for supplying the liquid chemical from the liquid chemical tank, a back pressure valve (BPV) **1120** mounted on the main line **1110** in front of the plurality of distribution suppliers **1200** with respect to a direction of the liquid chemical flowing through the main line **1110**, to maintain a pressure of the liquid chemical flowing in the main line **1110** at a preset certain main pressure, and a pressure sensor **1130** mounted on the main line **1110** between the BPV **1120** and the plurality of distribution suppliers **1200** to sense the pressure of the liquid chemical flowing in the main line **1110** behind the BPV **1120**.

(12) For example, the BPV **1120** may serve to control the pressure of the liquid chemical flowing in the main line **1110** to be maintained at the main pressure at a front end of the main line **1110**, and also serve to prevent backflow and oversupply of the liquid chemical through the main line **1110**.

(13) The plurality of distribution suppliers **1200** may be mounted at certain intervals along the main line **1110** of the main supplier **1100** to distribute the liquid chemical supplied through the main supplier **1100**, to the plurality of substrate processing apparatuses **10**.

(14) For example, the plurality of distribution suppliers **1200** may include a first distribution supplier **1200-1** for connecting between the main line **1110** and a first substrate processing apparatus **10-1**, a second distribution supplier **1200-2** for connecting between the main line **1110** and a second substrate processing apparatus **10-2** behind the first distribution supplier **1200-1** with respect to the direction of the liquid chemical flowing through the main line **1110**, a third distribution supplier **1200-3** for connecting between the main line **1110** and a third substrate processing apparatus **10-3** behind the second distribution supplier **1200-2** with respect to the direction of the liquid chemical flowing through the main line **1110**, an (n−1)th distribution supplier **1200-**$n-1$ for connecting between the main line **1110** and an (n−1)th substrate processing apparatus **10-4** behind the third distribution supplier **1200-3** with respect to the direction of the liquid chemical flowing through the main line **1110**, and an nth distribution supplier **1200-**n for connecting between the main line **1110** and an nth substrate processing apparatus **10-5** behind the (n−1)th distribution supplier **1200-**$n-1$ with respect to the direction of the liquid chemical flowing through the main line **1110**. Herein, although five distribution suppliers **1200** are branched from the main line **1110** in FIG. **1**, the number of distribution suppliers **1200** is not limited thereto and may include a wide variety of numbers depending on the number of substrate processing apparatuses **10** to be supplied with the liquid chemical.

(15) Although the configuration of only the first distribution supplier **1200-1** is representatively illustrated in FIG. **1** for convenience sake, the first distribution supplier **1200-1**, the second distribution supplier **1200-2**, the third distribution supplier **1200-3**, the (n−1)th distribution supplier **1200-**$n-1$, and the nth distribution supplier **1200-**n may equally include a distribution line **1210**, a first circulation line **1220**, and a second circulation line **1230**, which will be described below.

(16) The distribution line **1210** and the first and second circulation lines **1220** and **1230**, which may be included in each distribution supplier **1200**, will now be described in detail.

(17) As shown in FIG. **1**, the distribution supplier **1200** may mainly include the distribution line **1210** and the first and second circulation lines **1220** and **1230**. The distribution line **1210** may be branched from the main line **1110** of the main supplier **1100** and connect between the main line **1110** and each substrate processing apparatus **10**.
(18) Specifically, as shown in FIGS. **1** and **2**, a flow control valve (FCV) **1211** may be mounted on the distribution line **1210** to control a flow rate of the liquid chemical supplied to the substrate processing apparatus **10**. The FCV **1211** may determine a required flow rate of the liquid chemical in the distribution line **1210**. That is, the FCV **1211** may control the flow rate of the liquid chemical in the distribution line **1210**.
(19) A first on/off valve **1214** may be mounted on the distribution line **1210** behind the FCV **1211** with respect to a direction of the liquid chemical flowing through the distribution line **1210**, to open or close the distribution line **1210**.
(20) A flowmeter **1212** may be mounted on the distribution line **1210** in front of the FCV **1211** with respect to the direction of the liquid chemical flowing through the distribution line **1210**, to measure a flow rate of the liquid chemical flowing in the distribution line **1210**. A flow controller **1213** electrically connected to the FCV **1211** and the flowmeter **1212** may be mounted on the distribution line **1210** to receive a value of sensing the flow rate of the liquid chemical from the flowmeter **1212** and apply a control signal to the FCV **1211** based on the sensed value.
(21) For example, the flow controller **1213** may apply the control signal to the FCV **1211** based on a difference between a preset reference flow rate and an actual flow rate of the liquid chemical calculated using the sensed value received from the flowmeter **1212**, to control the difference between the reference flow rate and the actual flow rate of the liquid chemical within a preset certain error range.
(22) In this case, because the liquid chemical is supplied simultaneously to the plurality of substrate processing apparatuses **10** through the main supplier **1100** and thus the pressure for supplying the liquid chemical distributed through the main line **1110** of the main supplier **1100** to the distribution line **1210** of the distribution supplier **1200** is reduced, when an opening ratio of the FCV **1211** is rapidly increased to maintain an ejection rate of the liquid chemical supplied to each substrate processing apparatus **10**, a dead zone in the FCV **1211** may be reduced and thus particles stagnant in the FCV **1211** may be ejected together with the liquid chemical.
(23) However, because the distribution supplier **1200** according to the present invention includes the first and second circulation lines **1220** and **1230**, even when the liquid chemical is supplied simultaneously to the plurality of substrate processing apparatuses **10** through the main supplier **1100**, the pressure for supplying the liquid chemical distributed through the main line **1110** of the main supplier **1100** to the distribution line **1210** of the distribution supplier **1200** may be constantly maintained and thus the above-described problem may be solved.
(24) For example, as shown in FIG. **1**, the first circulation line **1220** may be branched from the distribution line **1210** and connect between the distribution line **1210** and the main line **1110** to circulate at least a portion of the liquid chemical flowing in the distribution line **1210** back to the main line **1110**.
(25) Specifically, the first circulation line **1220** may be branched from the distribution line **1210** behind the FCV **1211** with respect to the direction of the liquid chemical flowing through the distribution line **1210**, e.g., between the FCV **1211** and the first on/off valve **1214**, and connected to the main line **1110** in front of the BPV **1120** with respect to the direction of the liquid chemical flowing through the main line **1110**. A second on/off valve **1221** may be mounted on the first circulation line **1220** to selectively open or close the first circulation line **1220**.
(26) As such, the first circulation line **1220** may circulate the entirety of the liquid chemical supplied through the distribution line **1210** back to a front end of the BPV **1120** of the main line **1110** when the first on/off valve **1214** of the distribution line **1210** is closed, or circulate a portion of the liquid chemical supplied through the distribution line **1210** back to the front end of the BPV

1120 of the main line **1110** based on the pressure of the liquid chemical when the first on/off valve **1214** is open to supply the liquid chemical to the substrate processing apparatus **10**, thereby constantly maintaining the pressure of the liquid chemical flowing in the distribution line **1210**.
(27) As shown in FIG. **1**, the second circulation line **1230** may be branched from the distribution line **1210** in front of the first circulation line **1220** with respect to the direction of the liquid chemical flowing through the distribution line **1210**, and connect between the distribution line **1210** and the main line **1110** to circulate at least a portion of the liquid chemical flowing in the distribution line **1210** back to the main line **1110** in front of the first circulation line **1220**.
(28) Specifically, the second circulation line **1230** may be branched from the distribution line **1210** in front of the FCV **1211** and the flowmeter **1212** with respect to the direction of the liquid chemical flowing through the distribution line **1210**, and connected to the main line **1110** in front of the BPV **1120** with respect to the direction of the liquid chemical flowing through the main line **1110**.
(29) A relief valve **1231** may be mounted at a position where the second circulation line **1230** is branched from the distribution line **1210**, to circulate at least a portion of the liquid chemical flowing in the distribution line **1210** back to the main line **1110** when the pressure of the liquid chemical flowing in the distribution line **1210** exceeds a preset certain distribution pressure. A third on/off valve **1232** may also be mounted on the second circulation line **1230** to selectively open or close the second circulation line **1230**.
(30) At this time, in order to always bypass a certain amount of the liquid chemical to the second circulation line **1230** through the relief valve **1231**, the main supplier **1100** may set the main pressure of the liquid chemical having passed through the BPV **1120** and flowing in the main line **1110**, to be higher than the distribution pressure of the liquid chemical distributed through the distribution supplier **1200** and supplied to the substrate processing apparatus **10**.
(31) As such, by setting the main pressure of the liquid chemical flowing in the main line **1110** to be always higher than the distribution pressure required to appropriately supply the liquid chemical to the substrate processing apparatus **10** and by always bypassing a certain amount of the liquid chemical through the relief valve **1231** based on a difference between the distribution pressure and the main pressure, even when the liquid chemical is supplied simultaneously to the plurality of substrate processing apparatuses **10** through the main supplier **1100**, the pressure for supplying the liquid chemical distributed through the main line **1110** of the main supplier **1100** to the distribution line **1210** of the distribution supplier **1200** may be prevented from being reduced below the distribution pressure and thus changes in the opening ratio of the FCV **1211** to maintain the ejection rate of the liquid chemical supplied to the substrate processing apparatus **10** may be minimized.
(32) The substrate processing apparatus **10** including the above-described liquid chemical supply module **1000** will now be described in detail.
(33) Initially, as shown in FIG. **3**, the substrate processing apparatus **10** including the liquid chemical supply module **1000** according to an embodiment of the present invention may mainly include a chamber **100**, a gas supply unit **200**, a fan filter unit **300**, a substrate support unit **400**, an ejection unit **500**, a collection unit **600**, and a lift unit **700**.
(34) As shown in FIG. **3**, the chamber **100** may include a processing space A where a substrate S is cleaned. For example, the chamber **100** may have a rectangular, polygonal, or circular vessel shape to provide therein the processing space A where the substrate S is cleaned.
(35) The chamber **100** may include a gate **110** provided on a surface thereof facing a transfer chamber (not shown), and the gate **110** may be opened or closed by a door **120**. The gate **110** may function as an entrance through which the substrate S may enter.
(36) As shown in FIG. **3**, the substrate support unit **400** may be a kind of spin head and be mounted in the processing space A inside the chamber **100** to support and rotate the substrate S during the cleaning process.
(37) Specifically, the substrate support unit **400** may include a spin body **410**, support pins **420**,

chuck pins **430**, and a support shaft **440**, and the spin body **410** may have a substantially circular upper surface when viewed from above. The support shaft **440** rotatably driven by a driving means **450** may be fixed and coupled to a lower surface of the spin body **410**.

(38) A plurality of support pins **420** of the substrate support unit **400** may be provided on the upper surface of the spin body **410**. For example, a plurality of support pins **420** may be arranged at certain intervals along an edge of the upper surface of the spin body **410** and thus provided radially and equiangularly from a central axis (or rotational axis) of the spin body **410**. As described above, the support pins **420** may be arranged in a circular ring shape overall to support an edge of a rear surface of the substrate S such that the substrate S is spaced apart from the upper surface of the spin body **410** by a certain distance.

(39) A plurality of chuck pins **430** of the substrate support unit **400** may be provided on the upper surface of the spin body **410** and disposed farther away from the central axis of the spin body **410** than the support pins **420**. These chuck pins **430** may protrude more than the support pins **420** from the upper surface of the spin body **410** to support a side surface of the substrate S to prevent lateral dislocation of the substrate S due to centrifugal force when the spin body **410** rotates.

(40) The chuck pins **430** may be mounted to linearly move between a standby position and a support position which are spaced apart from each other by a certain distance along a radial direction of the spin body **410**. For example, the standby position may be a position farther from the central axis of the spin body **410** than the support position. As such, the chuck pins **430** may be positioned at the standby position when the substrate S is loaded onto or unloaded from the spin body **410**, and linearly move and be positioned at the support position when the substrate S is cleaned. At the support position, the chuck pins **430** may be in contact with the side surface of the substrate S.

(41) As shown in FIG. **3**, the collection unit **600** may be provided in a vessel shape with an open top to surround the substrate support unit **400** and mounted in the processing space A to collect a liquid chemical scattered from the substrate S when the substrate S rotates.

(42) For example, the collection unit **600** may have a vessel shape with an open top overall and include an inner collection barrel **610**, a middle collection barrel **620**, and an outer collection barrel **630**. The collection barrels **610**, **620**, and **630** may collect different liquid chemicals used for the cleaning process.

(43) Specifically, the inner collection barrel **610** may be provided in a circular ring shape surrounding the substrate support unit **400**, the middle collection barrel **620** may be provided in a circular ring shape surrounding the inner collection barrel **610**, and the outer collection barrel **630** may be provided in a circular ring shape surrounding the middle collection barrel **620**. A first space **610***a* of the inner collection barrel **610**, a second space **620***a* of the middle collection barrel **620**, and a third space **630***a* of the outer collection barrel **630** may function as inlets through which the liquid chemicals flow into the inner collection barrel **610**, the middle collection barrel **620**, and the outer collection barrel **630**, respectively.

(44) For example, the inlets of the collection barrels **610**, **620**, and **630** may be positioned at different heights, and collection lines **610***b*, **620***b*, and **630***b* may be connected to bottom surfaces of the collection barrels **610**, **620**, and **630**, respectively. The collection lines **610***b*, **620***b*, and **630***b* may discharge the liquid chemicals introduced through the collection barrels **610**, **620**, and **630**, respectively, and the discharged liquid chemicals may be recycled and reused through an external liquid chemical recycling system (not shown).

(45) As shown in FIG. **3**, the lift unit **700** may lift or lower the collection unit **600** in a vertical direction. When the collection unit **600** is lifted or lowered by the lift unit **700**, a height of the collection unit **600** relative to the substrate support unit **400** may be changed.

(46) The lift unit **700** may include a bracket **710**, a moving shaft **720**, and a h **730**. For example, the bracket **710** may be fixed and mounted on an outer wall of the collection unit **600**, and the moving shaft **720** moved in a vertical direction by the driver **730** may be fixed and coupled to the bracket

710.
(47) Specifically, when the substrate S is loaded onto or unloaded from the substrate support unit **400**, the collection unit **600** may be lowered by the lift unit **700** such that the substrate support unit **400** may protrude upward from the collection unit **600**. When the cleaning process is performed, the height of the collection unit **600** may be controlled based on the type of a liquid chemical supplied to the substrate S such that the liquid chemical may flow into a preset collection barrel **610**, **620**, or **630**.
(48) As shown in FIG. **3**, the gas supply unit **200** may supply air to the processing space A of the chamber **100**. For example, the gas supply unit **200** may supply temperature- and humidity-controlled air to the processing space A of the chamber **100**. To this end, although not shown in the drawings, the gas supply unit **200** may include a gas receiver provided with a filter to receive impurity-removed gas from the outside, a buffer for temporarily storing the received gas, a humidity controller for controlling a humidity of the gas, and a temperature controller for controlling a temperature of the gas.
(49) As shown in FIG. **3**, the fan filter unit **300** may be mounted on the chamber **100** to blow the air supplied from the gas supply unit **200**, into the processing space A. Specifically, the fan filter unit **300** may include a blower fan **310** for generating blowing force to blow the air into the processing space A, and a perforated plate **320** including a plurality of holes and provided in a flat plate shape having an area corresponding to a range of rotation of the blower fan **310**. In this case, the plurality of holes may be uniformly provided at regular intervals in the perforated plate **320** to induce the air supplied from the gas supply unit **200** to be uniformly spread and blown into the processing space A of the chamber **100**.
(50) As shown in FIG. **3**, the ejection unit **500** may be mounted above the substrate support unit **400** in the processing space A to eject at least one type of liquid chemical onto the substrate S through an ejection nozzle **510**.
(51) For example, the ejection unit **500** may receive a plurality of liquid chemicals from the above-described distribution supplier **1200** of the liquid chemical supply module **1000** according to an embodiment of the present invention and eject the liquid chemicals onto the substrate S through the ejection nozzle **510** provided with an ejection channel **511** in which the liquid chemicals flow. Although not shown in the drawings, a plurality of ejection units **500** may be provided. The ejection unit **500** may mainly include the ejection nozzle **510**, a support bar **520**, and a support shaft **530**.
(52) For example, the support shaft **530** may be disposed at a side of the collection unit **600** and rotated or lifted by a driving member **540**. The support bar **520** may have an end coupled to the support shaft **530** to support the ejection nozzle **510** mounted at another end thereof.
(53) The ejection nozzle **510** may be swung above the substrate support unit **400** by the rotation of the support shaft **530**. Herein, the liquid chemical supplied from the above-described liquid chemical supply module **1000** at a constant flow rate and ejected through the ejection nozzle **510** may be a chemical solution, a rinsing liquid, or a liquid organic solvent, and the flow rate for supplying the liquid chemical may be controlled by the liquid chemical supply module **1000**.
(54) Therefore, based on the liquid chemical supply module **1000** according to an embodiment of the present invention and the substrate processing apparatus **10** including the same, because the relief valve **1231** is mounted at a front end of the distribution line **1210** of the distribution supplier **1200** branched from the main line **1110** of the main supplier **1100** and connected to each substrate processing apparatus **10**, a pressure of the liquid chemical may be always constantly maintained by bypassing the liquid chemical when the pressure of the liquid chemical is higher than or equal to a set pressure.
(55) Accordingly, particle performance may be improved and equipment and chamber performance deviations between the substrate processing apparatuses **10** may be prevented by constantly maintaining a front pressure of the distribution line **1210** connected to each substrate processing

apparatus **10** and minimizing changes in the opening ratio of the FCV **1211** mounted at a rear end of the distribution line **1210** even when a plurality of substrate processing apparatuses **10** eject the liquid chemical.

(56) According to the afore-described embodiments of the present invention, because a relief valve is mounted at a front end of a distribution line of a distribution supplier branched from a main line of a main supplier and connected to each substrate processing apparatus, a pressure of a liquid chemical may be always constantly maintained by bypassing the liquid chemical when the pressure of the liquid chemical is higher than or equal to a set pressure.

(57) As such, a liquid chemical supply module capable of improving particle performance and preventing equipment and chamber performance deviations between substrate processing apparatuses by constantly maintaining a front pressure of a distribution line connected to each substrate processing apparatus and minimizing changes in an opening ratio of a FCV mounted at a rear end of the distribution line even when a plurality of substrate processing apparatuses eject a liquid chemical, and a substrate processing apparatus including the liquid chemical supply module may be implemented. However, the scope of the present invention is not limited to the above effects.

(58) While the present invention has been particularly shown and described with reference to embodiments thereof, it will be understood by one of ordinary skill in the art that various changes in form and details may be made therein without departing from the scope of the present invention as defined by the following claims.

Claims

1. A liquid chemical supply module comprising: a main supplier for supplying a liquid chemical from a liquid chemical tank to at least one substrate processing apparatus; and at least one distribution supplier for connecting between the main supplier and any one substrate processing apparatus to distribute the liquid chemical supplied through the main supplier to the any one substrate processing apparatus, wherein the distribution supplier comprises: a distribution line branched from a main line of the main supplier and connecting between the main line and the any one substrate processing apparatus; a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line; and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line.
2. The liquid chemical supply module of claim 1, wherein the main supplier comprises: the main line for supplying the liquid chemical from the liquid chemical tank; and a back pressure valve (BPV) mounted on the main line in front of the distribution line with respect to a direction of the liquid chemical flowing through the main line, to maintain a pressure of the liquid chemical flowing in the main line at a preset certain main pressure.
3. The liquid chemical supply module of claim 2, wherein the main supplier further comprises a pressure sensor mounted on the main line between the BPV and the distribution line to sense the pressure of the liquid chemical flowing in the main line behind the BPV.
4. The liquid chemical supply module of claim 2, wherein the distribution line comprises a flow control valve (FCV) mounted on the distribution line to control a flow rate of the liquid chemical supplied to the any one substrate processing apparatus.
5. The liquid chemical supply module of claim 4, wherein the distribution line further comprises a first on/off valve mounted on the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to open or close the distribution line.

6. The liquid chemical supply module of claim 4, wherein the first circulation line is branched from the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line.
7. The liquid chemical supply module of claim 6, wherein the first circulation line comprises a second on/off valve mounted on the first circulation line to open or close the first circulation line.
8. The liquid chemical supply module of claim 6, wherein the second circulation line is branched from the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line.
9. The liquid chemical supply module of claim 8, wherein the second circulation line comprises a relief valve mounted at a position where the second circulation line is branched from the distribution line, to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line when the pressure of the liquid chemical flowing in the distribution line exceeds a preset certain distribution pressure.
10. The liquid chemical supply module of claim 9, wherein the second circulation line further comprises a third on/off valve mounted on the second circulation line to open or close the second circulation line.
11. The liquid chemical supply module of claim 4, wherein the distribution line further comprises a flowmeter mounted on the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to measure a flow rate of the liquid chemical flowing in the distribution line.
12. The liquid chemical supply module of claim 11, wherein the distribution line further comprises a flow controller electrically connected to the FCV and the flowmeter to receive a value of sensing the flow rate of the liquid chemical from the flowmeter and apply a control signal to the FCV.
13. The liquid chemical supply module of claim 12, wherein the flow controller applies the control signal to the FCV based on a difference between a preset reference flow rate and an actual flow rate of the liquid chemical calculated using the sensed value received from the flowmeter, to control the difference between the reference flow rate and the actual flow rate of the liquid chemical within a preset certain error range.
14. The liquid chemical supply module of claim 2, wherein the main supplier sets the main pressure of the liquid chemical having passed through the BPV and flowing in the main line, to be higher than a distribution pressure of the liquid chemical distributed through the distribution supplier and supplied to the any one substrate processing apparatus.
15. The liquid chemical supply module of claim 1, wherein the distribution supplier comprises a plurality of distribution suppliers mounted at certain intervals along the main line of the main supplier to distribute the liquid chemical supplied through the main supplier, to a plurality of substrate processing apparatuses.
16. The liquid chemical supply module of claim 15, wherein the distribution suppliers comprise: a first distribution supplier for connecting between the main line and a first substrate processing apparatus; and a second distribution supplier for connecting between the main line and a second substrate processing apparatus behind the first distribution supplier with respect to a direction of the liquid chemical flowing through the main line.
17. The liquid chemical supply module of claim 16, wherein each of the first and second distribution suppliers comprises the distribution line, the first circulation line, and the second circulation line.
18. The liquid chemical supply module of claim 16, wherein the distribution suppliers further comprise an nth distribution supplier for connecting between the main line and an nth substrate processing apparatus behind an (n−1)th distribution supplier with respect to the direction of the liquid chemical flowing through the main line.

19. A substrate processing apparatus comprising: a chamber comprising a processing space where a substrate is processed; a substrate support unit mounted in the processing space to support and rotate the substrate; an ejection unit mounted above the substrate support unit in the processing space to eject at least one type of liquid chemical onto the substrate through an ejection nozzle; a liquid chemical supply module for supplying the liquid chemical to the ejection unit; and a collection unit provided in a vessel shape with an open top to surround the substrate support unit and mounted in the processing space to collect the liquid chemical scattered from the substrate when the substrate rotates, wherein the liquid chemical supply module comprises: a main supplier for supplying the liquid chemical from a liquid chemical tank to the ejection nozzle of the ejection unit; and a distribution supplier for connecting between the main supplier and the ejection nozzle to distribute the liquid chemical supplied through the main supplier to the ejection nozzle of the ejection unit, and wherein the distribution supplier comprises: a distribution line branched from a main line of the main supplier and connecting between the main line and the ejection nozzle; a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line; and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line.
20. A liquid chemical supply module comprising: a main supplier for supplying a liquid chemical from a liquid chemical tank to at least one substrate processing apparatus; and at least one distribution supplier for connecting between the main supplier and any one substrate processing apparatus to distribute the liquid chemical supplied through the main supplier to the any one substrate processing apparatus, wherein the distribution supplier comprises: a distribution line branched from a main line of the main supplier and connecting between the main line and the any one substrate processing apparatus; a first circulation line branched from the distribution line and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line; and a second circulation line branched from the distribution line in front of the first circulation line with respect to a direction of the liquid chemical flowing through the distribution line, and connecting between the distribution line and the main line to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line in front of the first circulation line, wherein the main supplier comprises: the main line for supplying the liquid chemical from the liquid chemical tank; a back pressure valve (BPV) mounted on the main line in front of the distribution line with respect to a direction of the liquid chemical flowing through the main line, to maintain a pressure of the liquid chemical flowing in the main line at a preset certain main pressure; and a pressure sensor mounted on the main line between the BPV and the distribution line to sense the pressure of the liquid chemical flowing in the main line behind the BPV, wherein the distribution line comprises: a flow control valve (FCV) mounted on the distribution line to control a flow rate of the liquid chemical supplied to the any one substrate processing apparatus; a flowmeter mounted on the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to measure a flow rate of the liquid chemical flowing in the distribution line; and a first on/off valve mounted on the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, to open or close the distribution line, wherein the first circulation line is branched from the distribution line behind the FCV with respect to the direction of the liquid chemical flowing through the distribution line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line, and wherein the second circulation line is branched from the distribution line in front of the FCV with respect to the direction of the liquid chemical flowing through the distribution

line, and connected to the main line in front of the BPV with respect to the direction of the liquid chemical flowing through the main line, and comprises a relief valve mounted at a position where the second circulation line is branched from the distribution line, to circulate at least a portion of the liquid chemical flowing in the distribution line back to the main line when the pressure of the liquid chemical flowing in the distribution line exceeds a preset certain distribution pressure.